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MACHINE DYNAMICS: A LITERATURE REVIEW

A Literature Review of Gas Turbine Blade Dynamic Modeling and Vibrations

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Abstract

Gas turbine blade operates under cyclic thermo-mechanical loading and are susceptible to high-cycle fatigue driven by vibratory stresses. Turbine blades are one of the critical components that fail due to self-excited vibrations caused by the wake of neighboring blades or unsteady inlet flow. The service life of a turbine blade is often limited by creep, low-cycle fatigue (LCF), high-cycle fatigue (HCF), and surface attack. As vibration can cause severe damage to rotating blades and pose a safety hazard in the event of failure, it is mandatory to design the turbine blades effectively and minimize vibration excitation and dynamically unsafe operating conditions. This work reviews the design of turbine blades with finite element (FE) modeling, considering geometrical, material, and nonlinear modeling. Vibration analysis techniques, including frequency analysis, blade-tip-timing, nonlinear vibrational analysis, and neural networks, are also reviewed. Vibration control with damping, mistuning, and excitation reduction is reviewed in the section on vibration control methods.

Keywords: Turbine blade, vibration, vibration control, damping, blade tuning, nonlinear modeling, frequency analysis

1. Introduction

The design theory behind a turbine blade in a turbomachinery is complex and exhibits an infinite number of vibration modes at high rotational speeds, steep thermal gradients, and complex unsteady aerodynamic loading. Temperature can change stiffness by altering the material modulus [1]. Natural frequencies also increased with high shaft speed due to centrifugal stiffening. Due to a significant temperature rise, the turbine's natural frequencies generally fall with increasing shaft speed. These conditions can induce significant vibratory stresses and lead to resonance, high-cycle fatigue, cracks, and, in a worst-case scenario, catastrophic failure [2].

In gas turbines, the two principal sources of blade vibrations are the excitation of blade vibrations caused by time-varying forces acting on the rotating turbine blades and the excitation of blade vibrations induced from the coupling effect between the blade vibrations and torsional turbine vibrations [3]. When these effects accumulate over time, excitations will eventually generate a reaction throughout the whole system and become the system response.

The typical blade dynamic design approach involves an "M4 approach", that is, minimize the force applied to the rotor, minimize exposure to potentially damaging resonances, maximize the source of damping, and maximize the available high-cycle fatigue strength [4], [5].

The purpose of this literature review is to review how turbine blades are modelled, what the vibration analysis techniques are, and how to control the vibration induced

Dynamic Blade Modeling

Gas turbines are widely used in power generation, industrial, aviation, and marine applications [6]. Precise modeling is necessary to predict the vibrational behavior of turbine blades during operation accurately.

Finite element (FE) modeling is a well-known and relatively mature approach for modeling the dynamic behavior of gas-turbine blades, because it provides the fundamental mass-stiffness representation needed for modal and forced-response analyses:

$$f_e = K_e u_e \tag{1}$$

where f_e is the element nodal force vector, K_e is the element stiffness matrix, and u_e is the element nodal displacement vector [7]. Before performing FE modeling, the blade geometry and kinematics, material properties, boundary conditions, contact interfaces, loads, analysis types, and element mesh types must be defined [8]. 3D solid or shell elements are used for mesh generation, depending on the computational cost and structural complexity [8]. Thermal boundary conditions, such as temperature distributions, and mechanical boundary conditions, such as load, contact, and movement, are also accounted for in FE calculations [5]. Nonlinearity during turbine operation is modeled through iterative calculations, in which geometric and material nonlinearity, such as plasticity and creep, greatly influence the performance of turbine blades [8]. Modal analysis is used to determine the natural frequencies and mode shapes, including bending, torsion, coupled modes, which are then used as the basis for harmonic and/or transient analyses that simulate the forced response due to gas pressure excitation, unbalance, or shaft vibration [4], [8], [9].

In recent years, the most used commercial software for dynamic blade FE modeling in both research and industry includes Ansys (Ansys Inc.), Abaqus (Dassault Systèmes SIMULIA), MSC Nastran (Hexagon), and Simcenter STAR-CCM+ (Siemens Digital Industries Software).

1.1 Geometrical and Material Modeling

In a study of a single industrial gas turbine blade, Kumar and Pandey [6] developed a 3D FE model with appropriate root constraints. They subjected it to centrifugal and static loads to compute stress distributions, followed by an eigenvalue analysis to determine natural frequencies and mode shapes. The results showed that the expected sequence of bending modes, torsional modes, along with geometrical features such as fillets and profile thickness, and material choice, strongly influence local stiffness and mode localizations, which are essential for generating Campbell diagrams and assessing potential resonances [6], [9]. Figure 1 shows a representative Campbell diagram for a turbine blade, where the intersections between excitation lines and natural frequency curves indicate potential resonance.

Comparative studies have been carried out for decades to understand how different materials affect blade vibration and characteristics. In the work of Efe-Ononeme *et al.* [10], they assigned different elastic moduli and densities to the same blade geometry. FE modal analysis was used to compare two nickel-based superalloys, i.e., UDIMET 500 (U500) and IN738, to determine their natural frequencies and mode shapes [10]. Their results showed that U500 has a natural frequency of 896 Hz, while IN738 has 751 Hz under the same blade geometry and loading at an operational excitation frequency of 85 Hz, implying larger frequency separation and reduced resonance risk for U500 in that application. Building on such work, Singth *et al*. [11] presented a computational analysis of gas turbine blades with four different materials, i.e., titanium alloy, Nimonic 80A, Inconel 517, rhenium, to compare the modal frequencies and deformation patterns. Singth *et al*. [11] found that Nimonic 80A exhibits the highest modal frequencies and the lowest deformation level among the materials studied.

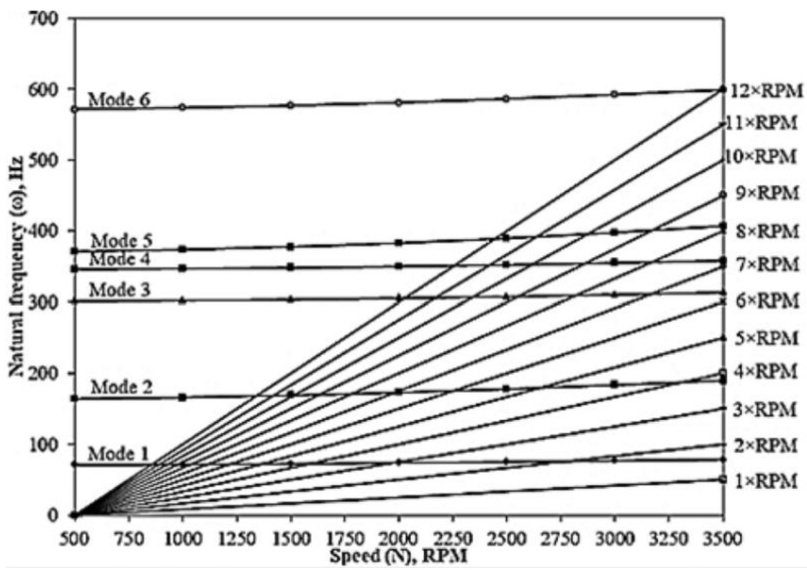

Figure 1: Example of a Campbell Diagram of a Turbine Blade [12]

1.2 Nonlinear Modeling

During FE modeling of large deformations and contact interactions, the turbine blade will show nonlinearities depending on whether we are analyzing geometric, material, or contact interface nonlinearity [8]. Hou *et al.* [13] modeled a turbine blade and disc assembly combining centrifugal, gas pressure, thermal expansion, and contact interactions. They investigated the nonlinear FE analysis considering geometric nonlinearity [13]. The stress field is computed in the deformed configuration and later used as the initial state for a pre-stressed modal analysis [13]. Their methods showed that, under operating conditions, the blade's natural frequencies increased by 10% compared to those from a slight-deflection clamped test carried out under static conditions at room temperature [13]. Wang *et al.* [14] presented a detailed nonlinear dynamic model of turbine runner blades with a crack for the treatment of geometrical nonlinearity. In the work of Wang *et al.* [14], large deformation is represented by a shell element with strain-displacement relations, in which quadratic terms in the displacement gradients are included. The crack stiffness matrix depends on material parameters, structural dimensions, and crack parameters, which are determined during the analysis of vibration and fatigue characteristics [14]. The effective stiffness becomes a nonlinear function of both geometry and material.

These contributions show that FE modeling is essential for obtaining turbine blade natural frequencies and mode shapes. From there, we can better understand the influence of blade design on vibration, optimize the design, formulate vibration control methods, and apply appropriate diagnostic techniques. Moreover, the aerodynamic, cooling, and mechanical design of turbine blades are also taken into account to maximize turbine efficiency, extend service life, and reduce production costs and weight [5], [15].

2. Blade Vibration Analysis Methods

Vibrations can cause severe damage to the mechanical components and pose potential safety hazards. It is absolutely necessary to avoid the excitation frequencies of the aerodynamic forces that match the range of the natural frequencies of the bladed disk [16]. Campbell diagrams are used to ensure that the operating speed does not excite frequencies that fall within the range of the rotating blade's natural frequency, thus avoiding resonance [3]. The growth of vibration resonance is governed by the magnitude of the excitation force, the damping of the blade material, and the resonance response factor [3]. Blade vibration can be categorized into forced and self-excited vibration, as shown in Figure 2**Error! Reference source not found.** [16], [17].

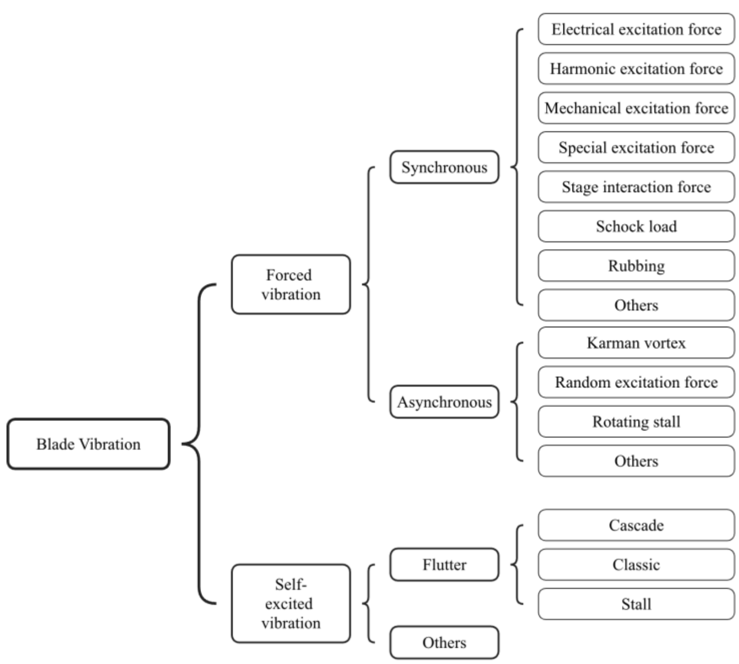

Figure 2: Blade vibration category

2.1 Frequency Analysis

Frequency spectrum analysis techniques are the most commonly used method for signal analysis and diagnosis [18]. The measured vibration signals are converted from the time domain to the frequency domain [18]. The power spectrum and the energy associated with specific critical frequencies are evaluated in the frequency domain [18]. Until today, variations in the vibration amplitudes of the blade passing frequency (BPF) and its harmonics are still frequently used as the primary detection measure for identifying and diagnosing blade faults [18]. Chu *et al.* [19] reported that narrow-band, high-resolution Fast-Fourier-Transformation (FFT) can be applied around critical frequencies for early detection, with characteristic sideband patterns providing fault fingerprints for rotating machinery.

2.2 Blade Tip Timing (BTT)

Bornassi *et al.* [20] proposed that blade tip-timing (BTT) can be used to measure blade vibration. To detect blade vibrations, Bornassi *et al.* [20] used eighteen high-temperature capacitive sensors at the last three stages of the turbine section. Bornassi *et al.* [20] converted the time-of-arrival deviations of the blade tips into blade tip deflection, then into frequency spectra. These BTT measurements are post-processed to extract the main resonance parameters of blade vibration (amplitude, frequency, damping, and phase) [20]. For each turbine stage, a Campbell diagram is generated, damping is evaluated as a function of rotational speed, and the respective resonance amplitudes are determined from these results [20]. Li *et al.* [21] developed an experimental BTT system that focuses on improving the hardware so that advanced under-sampled frequency analysis algorithms can be applied to blade vibration data accordingly. Li *et al.* [21] designed a high-speed rotor rig with low casing vibration to reach blade resonance and keep the prob motion small. Then, they integrate multiple probes and a once-per-revolution (OPR) sensor to obtain blade tip time-of-arrival signals. Li *et al.* [21] demonstrated that their experimental rig could apply BTT frequency-analysis methods to real rotating hardware and compare the extracted frequency results with finite element method (FEM) results, further demonstrating that the system can support practical blade vibration monitoring.

2.3 Nonlinear Vibration Analysis

In reality, when there are cracks, frictional contacts, rubbing, or large geometric deformation, linear analysis can no longer accurately represent the real-world dynamic behavior. In this case, nonlinear vibration analysis is required. Amplitude-dependent stiffness and non-harmonic responses are introduced in blade vibration dynamics analysis [22]. Nonlinear dynamic models of rotating blades with breathing cracks show that crack opening and closing modulate stiffness, lower effective resonant frequencies, and generate superharmonic resonances and complex response patterns that depend strongly on the crack position and depth [14], [23]. Shen *et al.* [24] conducted both theoretical and experimental harmonic analyses of crack blades. They demonstrated that crack-induced nonlinearities produce higher-harmonic components in the response spectra, which can be regarded as sensitive indicators for crack detection. In another study by Sanliturk *et al.* [25], a first-order harmonic balance method was applied to multi-degree-of-freedom blade models with friction dampers to predict resonance response in the frequency domain. Sanliturk *et al.* [25] reported that friction contact dynamics are nonlinear, which makes the time-domain solutions expensive. Thus, they presented a method, namely the frequency-domain first-order harmonic balance method (HBM), also described this as an equivalent linearization approach by representing the nonlinear friction interface as an amplitude-dependent complex stiffness. Then, they performed time-marching analyses using continuous sine sweep excitation and investigated the influence of different damping levels and the stiffness distributions on the response curve and resonance level.

2.4 Neural Networks

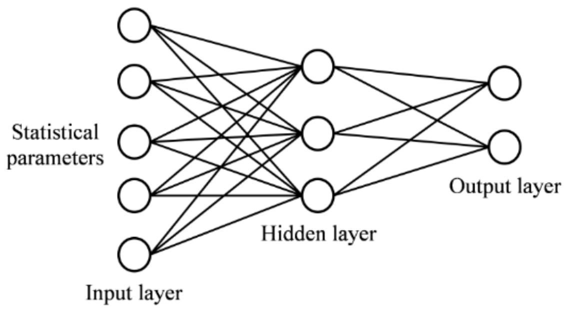

Figure 3: Overview of an ANN architecture by [25]

Neural networks are effective for vibration-based fault diagnosis of rotating blades, as conventional frequency-domain analysis, such as Fourier analysis, and time-frequency-domain analysis, such as wavelet analysis, require experienced interpretation and remain subjective [2]. Ngui *et al.* [2] proposed a method based on time-frequency features feature extraction, and the use of artificial intelligence (AI). The continuous wavelet transform (CWT) was applied to casing vibration signals, and wavelet coefficients were extracted at the operating frequency and blade-passing frequencies to compute statistical features [2]. Using these wavelet-derived statistical features, they trained a supervised artificial neural network (ANN) to classify blade conditions, which is a better approach than manual spectrum interpretation using FFT spectra or wavelet maps [2]. Their results showed that the ANN-trained method achieved the highest classification accuracy of 88.43% [2]. In comparison, the features extracted from the operating frequency and blade passing frequencies achieved 80.56% accuracy, whereas those extracted from the blade passing frequencies achieved only 64.58% [2].

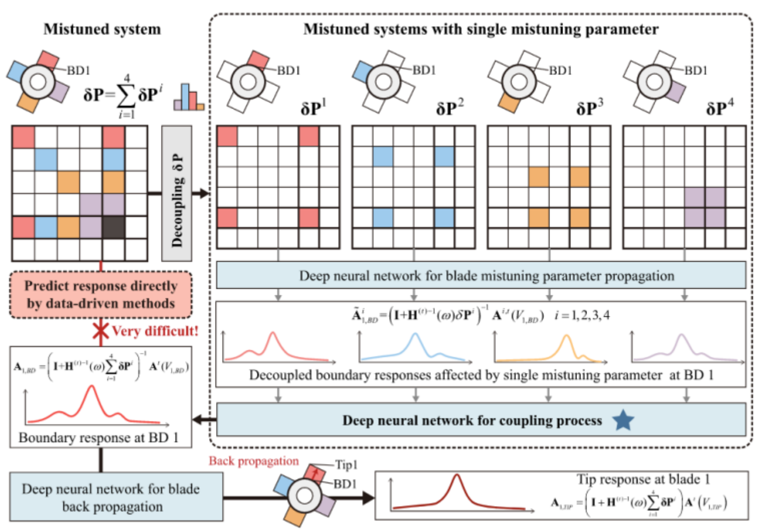

Figure 4: Schematic of boundary response decoupling and blade tip response prediction for a four-blade mistuned system [26]

Liang *et al.* [26] presented a specialized deep neural network framework for a mistuned system (MS-DNN) to predict the dynamic response of mistuned bladed disks from mistuning parameters for both the lumped parameter model and the large-scale FE model. MS-DNN was introduced at the blade level and disc level [26]. The blade-disk coupling calculations are replaced by using blade-level networks for parameter propagation and disc-level networks for interface coupling via boundary nodes [26]. Their proposed method decouples the system's vibration equation and replaces the corresponding coupling stage with a neural network [26]. Their results showed that all neural networks in MS-DNN showed high prediction accuracy on the training and unknown test sets [26].

3. Vibration Control Methods

Vibration control for turbine blades can be categorized into passive and active operational measures. This literature review focuses on discussions of passive design measures. In the following section, vibration control by using passive control with the use of friction damping, blade mistuning, and reduction of excitation is discussed.

3.1 Vibration Control by Damping

Friction damping is widely used through under-platform dampers, midspan dampers, shrouds, and snubbers. Damping in the blade system helps to reduce the blade vibrations, especially under resonance conditions.

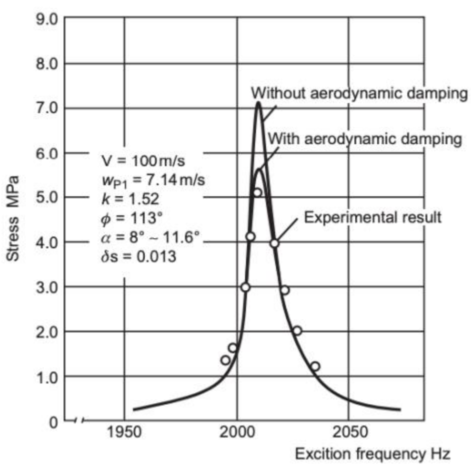

Figure 5: Vibration stress level in a blade

Under-platform dampers are small metallic elements inserted between two adjacent blade platforms [27]. The centrifugal force presses them into contact so that relative motion during vibration produces sliding or micro-slip [17], [28]. Wu *et al.* [28] presented their work, which used under-platform dampers and finite element forced-response analysis to analyse high-speed rotating bladed disks. Wu *et al.* [28] reported that the resonant amplitudes and dynamic stresses can be reduced to those of the undamped configuration if the damper mass, contact stiffness, and friction are tuned correctly. However, if the damper parameters are not selected correctly, the mode shapes will be changed significantly, and only a limited number of blades can localize the vibratory response [28].

Midspan dampers connect neighbouring blades at an intermediate span location and introduce amplitude-dependent stiffness and damping through frictional contacts. Ferhatoglu *et al.* [29] used midspan dampers to perform nonlinear vibration analysis of turbine-bladed disks. In Ferhatoglu *et al.*'s [29] work, midspan dampers exhibit the potential of flattening resonance peaks and can reduce the maximum steady state response. However, the use of midspan dampers may also induce jump and hysteresis phenomena in the frequency-response curves and should be taken into account when designing turbine blades [29].

3.2 Vibration Control by Blade Mistuning

Another method of counter-measuring vibration on the turbine blade and bladed-disc is blade mistuning [16]. If done with purpose, mistuning is called detuning [16]. Blade mistuning is the modification of blade geometry or stiffness to shift its natural frequencies away from known excitation frequencies [17]. Zinkovskyi *et al.* [30] presented that specific local mistuning patterns in aircraft compressors and turbine disks can either amplify or decrease vibration stresses, providing a basis for mistuning-aware tolerance specifications. Doligalski *et al.* [31] applied reduced-order models to assembly-defined manufacturing tolerances to obtain frequency scatter and mode localization patterns for 83 bladed-discs under static conditions. Doligalski *et al.* [31] showed that the bladed discs with a mistuning distribution that minimizes the forced response can reduce peak responses compared with both perfectly tuned and randomly mistuned configurations, thus avoiding rotating blades from high-cycle fatigue failure.

3.3 Vibration Control by Reduction of Excitation

One of the significant sources of excitation force is the unsteady aerodynamic flows. Several studies have used Computational Fluid Dynamics (CFD) software to optimize steam paths when designing turbine blades. The most used CFD software are Ansys CFX & Ansys Fluent (Ansys Inc.), OpenFOAM (OpenFOAM Ltd.), and COMSOL Multiphysics (COMSOL). Schwerdt *et al.* [32] used CFD to analyze 3D-scanned new and worn blades and to analyze the aerodynamic pressure and temperature, which act as the sources of excitation for vibration. They mapped these aerodynamic forces into a full bladed disk structural dynamics model, in which the change of mistuning sensitivity is also included, to evaluate forced response amplitude. Schwerdt *et al.* [32] showed that wear can significantly change the modal excitation and resonance response. Moreover, Schwerdt *et al.* [32] also reported that aerodynamic damping should be included in future investigations to understand the fluid-structure interaction better and its impact on vibration levels.

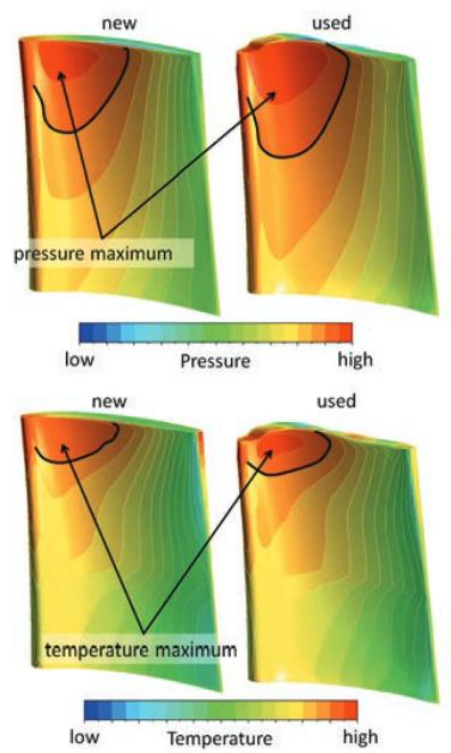

Figure 6: CFD simulation on new and used turbine blade with total pressure and temperature distribution by Schwerdt *et al.* [32]

4. Conclusion and future research outlook

Turbine blades are one of the parts in gas turbines that experience the highest loads during operation and are among the most expensive due to the advanced materials required and the complex geometry involved in production and repair. It is necessary to define the parameters and requirements in detail and understand the turbine blade dynamics thoroughly when designing turbine blades. Across the reviewed studies, blade geometry, material properties, realistic boundary conditions, and contact conditions strongly influence the localization of stiffness, mode shapes, and frequency separation from excitation sources. Recent neural network frameworks show potential for more accurate response prediction and for supporting classification tasks. Future research should prioritize high-fidelity coupled thermo-mechanical-aeroelastic modeling, improve friction, and improve contact parameter identification. With the widespread use of AI and ML in recent years, these technologies can be integrated into design optimization and applied to real-time risk assessment to improve data collection accuracy.

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